If the wire is kept in a magnetic field $\vec{B}$, then average force experienced by the charge (here, electron) in the wire is

$$\vec{F} = -e\left(\vec{v}_d \times \vec{B}\right)$$

Let n be the number of free electrons per unit volume, therefore

$$n = \frac{N}{V}$$

where N is the number of free electrons in the small element of volume V = A *dl*.

Hence Lorentz force on the wire of length d*l* is the product of the number of the electrons

(N = nA d*l*) and the force acting on an electron.

$$d\vec{F} = -enAdl\left(\vec{v}_d \times \vec{B}\right)$$

The length d*l* is along the length of the wire and hence the current element in the wire is $I\,d\vec{l} = -enA\vec{v}_d dl$ Therefore the force on the wire is

$$d\vec{F} = \left(I\,d\vec{l} \times \vec{B}\right) \qquad (3.65)$$

The force in a straight current carrying conducting wire of length l placed in a uniform magnetic field is

$$\vec{F} = \left(I\vec{l} \times \vec{B}\right) \qquad (3.66)$$

In magnitude,

$$F = BIl\sin\theta$$

(a) If the conductor is placed along the direction of the magnetic field, the angle between them is θ = 0°. Hence, the force experienced by the conductor is zero.

(b) If the conductor is placed perpendicular to the magnetic field, the angle between them is θ =90° Hence, the force experienced by the conductor is maximum, which is $F = BIl$.

Fleming's left hand rule (mnemonic)

When a current carrying conductor is placed in a magnetic field, the direction of the force experienced by it is given by Fleming's Left Hand Rule (FLHR) as shown in Figure 3.57.

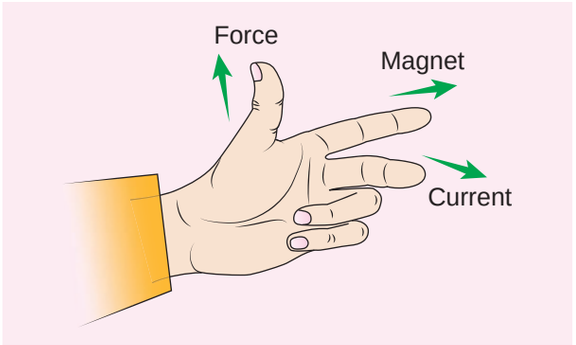
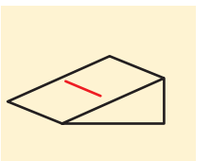


Figure 3.57 Fleming's Left Hand Rule (FLHR)

Stretch forefinger, the middle finger and the thumb of the left hand such that they are in mutually perpendicular directions. If forefinger points the direction of magnetic field, the middle finger points the direction of the electric current, then thumb will point the direction of the force experienced by the conductor.

EXAMPLE 3.27

A metallic rod of linear density is 0.25 kg m^{-1} is lying horizontally on a smooth inclined plane which makes an angle of 45° with the horizontal. The rod is not allowed to slide down by flowing a current through it when a magnetic field of strength 0.25 T is acting on it in the vertical direction. Calculate the electric current flowing in the rod to keep it stationary.



UNIT-3(XII-Physics_Vol-1).indd 184 04-03-2019 10:35:25